

Putting Context in Context: the Impact of Discussion Structure on Text Classification

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Introduction

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Online Conversations

Conversations on Social Media Platforms (SMPs) where people discuss



Text: what is written

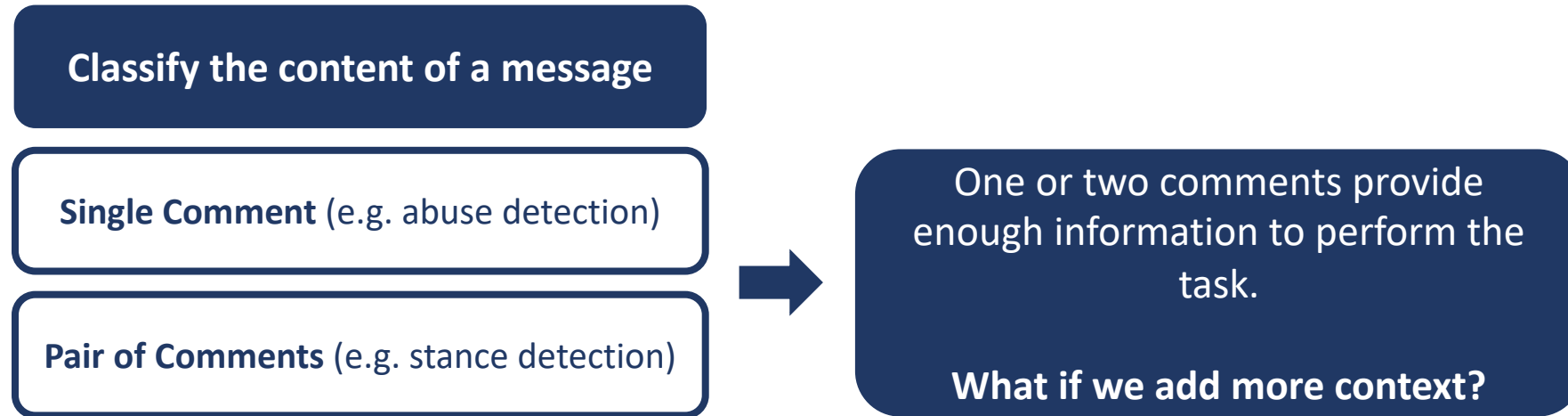


Time: when it was written



Structure: how people interact

Task: Text Classification



Task: Text Classification

Adding context is **not always**
beneficial

Menini et al. (2021)
Anuchitanukul et al. (2022)

Is contextual information beneficial?

And **when** it is beneficial?

Past works

Menini et al. (2021), Anuchitanukul et al. (2022), Yu et al. (2022): discussion around the **usefulness of (textual) context**.

Tian et al. (2022): combination of **transformer-based** and **GNN-based** models with user information on discussion trees.

Agarwal et al. (2022): Graph Walk on **discussion trees** and transformer-based model to encode sentences.

Yang et al. (2019): **text concatenation** for context.

Contributions

We integrate **textual**, **temporal** and **structural** context, expressing such information in **natural language** and exploiting a transformer-based model for a text classification task.

We avoid to explicitly provide user-related related information, representing them through “**local discussion IDs**”:

- **privacy-preserving**;
- **profiling-preserving**.

We perform our experiments on **three different datasets**, focusing on the **biggest** one and evaluating:

- the **learning curve**;
- performance on **more “complex” chains**.



Scenario

Discussion Chain



Text



User



Time

GOAL
Classify C_{12}



Inject
Contextual
Information

Inject Context

IDEA 1

Keep “local” user and time information

User: index in order of first appearance in the discussion



Time: time passed from the posting of the initial claim



IDEA 2

Design specific **input configurations**, reporting also **user** and **time** information in natural language

User: “ n th user”



Time: “after d days, h hours, m minutes”



Inject (only) Textual Context

SINGLE

`<s>` Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible. `</s>`

Inject (only) Textual Context

PAIR

<s> Personal "return on investment" should not be a guide on charity. </s>

</s> Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible. </s>

Inject (only) Textual Context

TC (Textual Context)

<s> People should donate to organisations that support gorillas instead of to those that support starving children. </s>

</s> Saving gorillas has less impact on the donor's own well-being than saving a child. </s>

</s> Personal "return on investment" should not be a guide on charity. </s>

</s> Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible. </s>

Inject Temporal and Structural Context

<t> after 28 days, 23 hours, 14 minutes </t>

<o> 1st user </o>

Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible.

Inject Temporal Context

TC + T (Textual Context + Time)

<s> <t> after 0 days, 0 hours, 0 minutes </t> People should donate to organisations that support gorillas instead of to those that support starving children. **</s>**

</s> <t> after 0 days, 6 hours, 36 minutes </t> Saving gorillas has less impact on the donor's own well-being than saving a child. **</s>**

</s> <t> after 7 days, 20 hours, 23 minutes </t> Personal "return on investment" should not be a guide on charity. **</s>**

</s> <t> after 28 days, 23 hours, 14 minutes </t> Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible. **</s>**

Inject Structural Context

TC + U (Textual Context + User)

<s> <o> 0th user </o> People should donate to organisations that support gorillas instead of to those that support starving children. </s>

</s> <o> 1st user </o> Saving gorillas has less impact on the donor's own well-being than saving a child. </s>

</s> <o> 2nd user </o> Personal "return on investment" should not be a guide on charity. </s>

</s> <o> 1st user </o> Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible. </s>

Inject Temporal and Structural Context

TC + U + T (Textual Context + User + Time)

<s> <t> after 0 days, 0 hours, 0 minutes </t> <o> 0th user </o> People should donate to organisations that support gorillas instead of to those that support starving children. **</s>**

</s> <t> after 0 days, 6 hours, 36 minutes </t> <o> 1st user </o> Saving gorillas has less impact on the donor's own well-being than saving a child. **</s>**

</s> <t> after 7 days, 20 hours, 23 minutes </t> <o> 2nd user </o> Personal "return on investment" should not be a guide on charity. **</s>**

</s> <t> after 28 days, 23 hours, 14 minutes </t> <o> 1st user </o> Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible. **</s>**

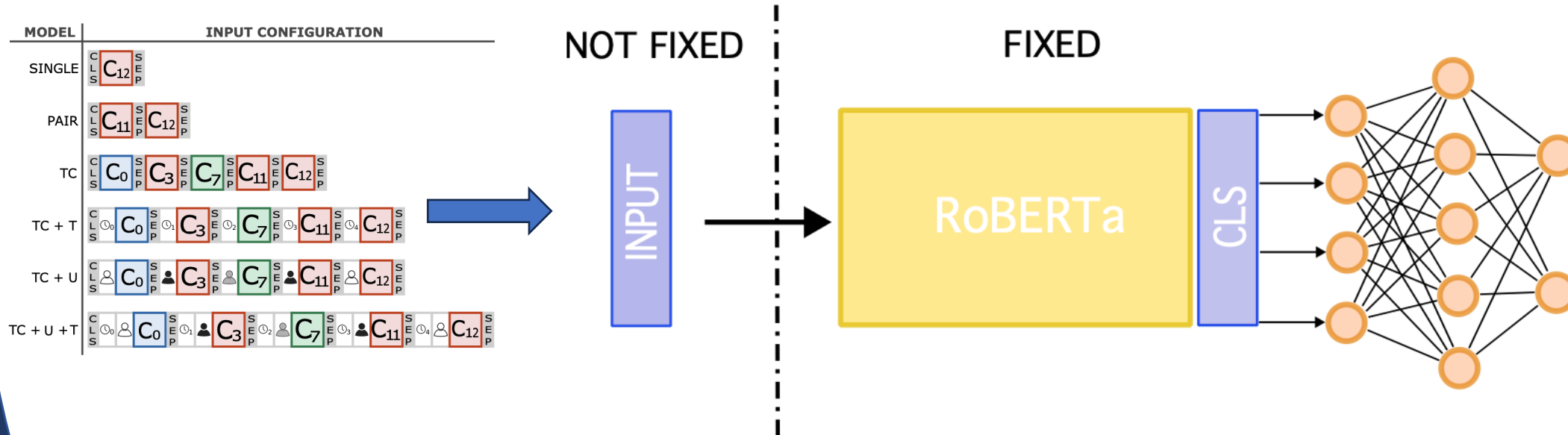
Experiments

Model



MODEL	INPUT CONFIGURATION
SINGLE	C L S C₁₂ S E P
PAIR	C L S C₁₁ S E P C₁₂ S E P
TC	C L S C₀ S E P C₃ S E P C₇ S E P C₁₁ S E P C₁₂ S E P
TC + T	C L S ⌚₀ C₀ S E P ⌚₁ C₃ S E P ⌚₂ C₇ S E P ⌚₃ C₁₁ S E P ⌚₄ C₁₂ S E P
TC + U	C L S ⧑ C₀ S E P ⧑ C₃ S E P ⧑ C₇ S E P ⧑ C₁₁ S E P ⧑ C₁₂ S E P
TC + U + T	C L S ⌚₀ ⧑ C₀ S E P ⌚₁ ⧑ C₃ S E P ⌚₂ ⧑ C₇ S E P ⌚₃ ⧑ C₁₁ S E P ⌚₄ ⧑ C₁₂ S E P

Model



Fine-tune pretrained RoBERTa model
(Liu et al., 2019) with MLP head

Kialo Dataset for Stance Detection

- Scialom et al. (2020)
- Discussion chains from 2 to 38 claims
- Binary Classification Task
- **Label:** stance of the last claim towards the previous one (Support vs Contrast)
- Slightly unbalanced label distribution
- **122k items** in training

Personal "return on investment" should not be a guide on charity.

Receiving a benefit while helping others is not morally wrong. Otherwise all the foundations supported by big brands would be morally reprehensible.

CONTRAST

Results on Kialo Dataset

Powerful Baseline: **PAIR** model

M-F1 score to compare
models' performance

OUTCOME

Structural Context is
extremely **effective!**

Category	Model	M-F1
BASELINES	PAIR	81.2 (± 0.2)
CONTEXTUAL	TC	80.5 (± 0.3)
	TC + T	81.7 (± 0.3)
	<i>TC + U</i>	<i>83.7 (± 0.5)[♦]</i>
	<i>TC + U + T</i>	<i>83.9 (± 0.3)[♦]</i>

♦ : significant improvement w.r.t. PAIR model
for **Student's t-test**

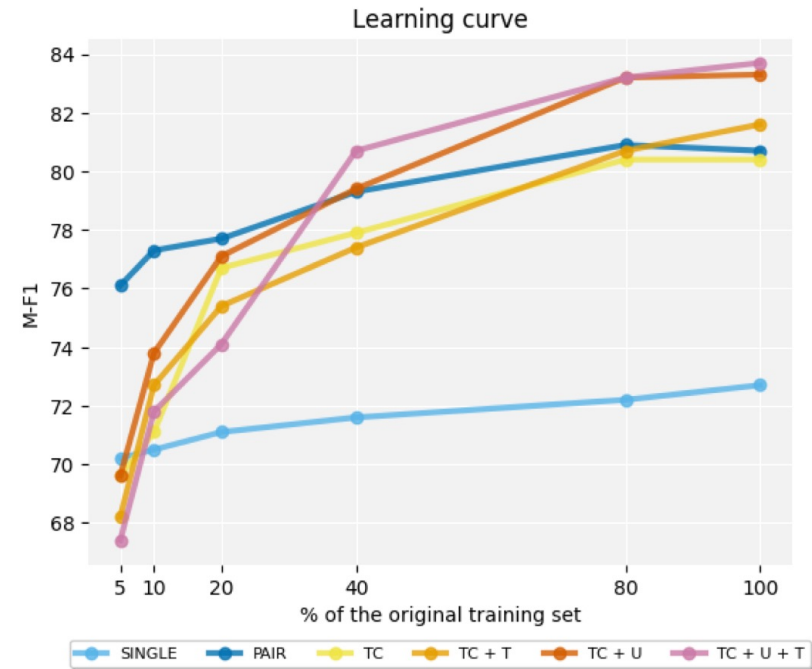
Learning Curve

Learning Curve

We analyzed the performance on the Kialo dataset at **varying training size**.

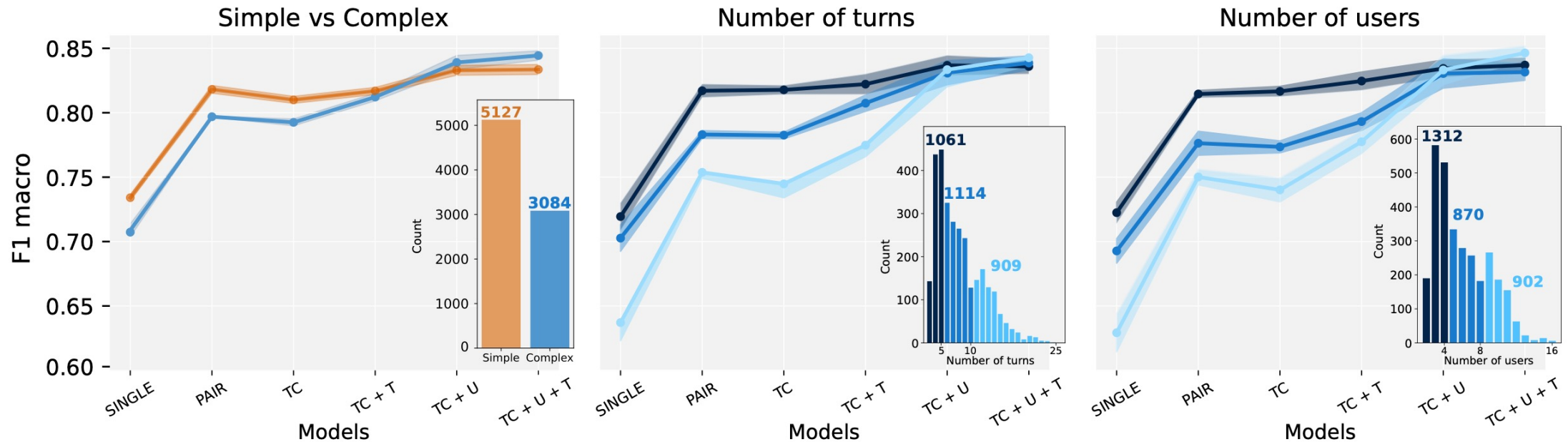
Original size: **122k**.

With increasing training data: **structural contextual models significantly outperform baselines**.



Analysis
of chain
complexity

Analysis of chain complexity



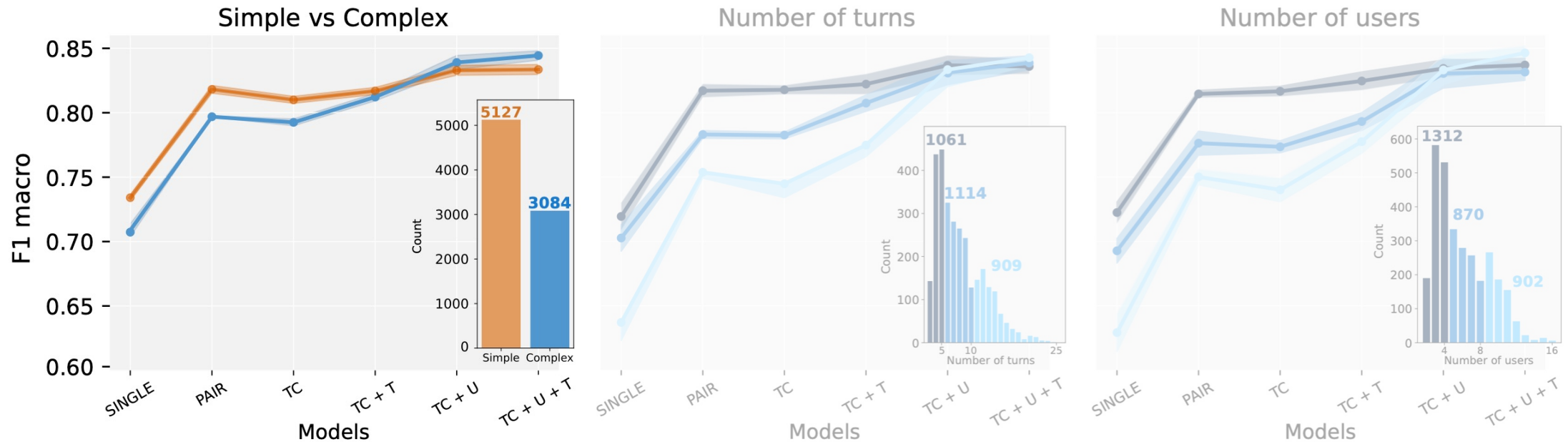
Consecutive claims from the same authors merged in a **single turn**.

Simple discussion chain: discussion where each user contributes with only one turn.

Complex discussion chain: discussion where at least one user has more than one turn.

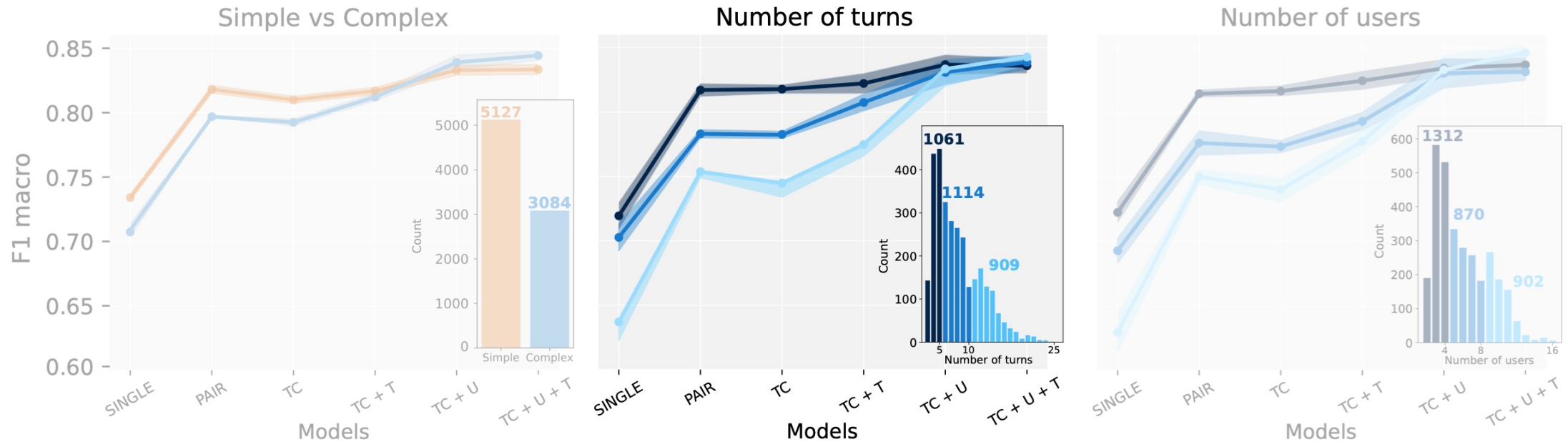
Results on Complex chains

Model with structural context: better performance on complex chains



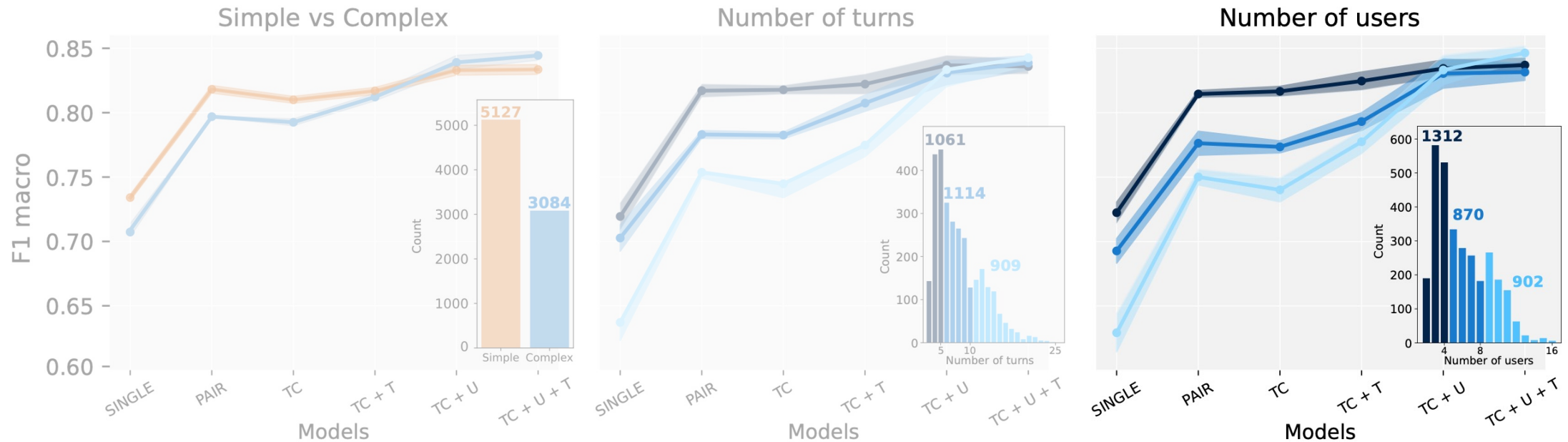
Results on Complex chains

Model with structural context: stable performance across different number of turns



Results on Complex chains

Model with structural context: stable performance across different number of users



Results on two Smaller Datasets...

Abusive Language Detection
dataset from Menini et al. (2021)

Retrieved from Twitter

5k items in training

Binary classification

Highly unbalanced

Stance Detection dataset from
Gorrell et al. (2019)

Retrieved from Twitter

4k items in training

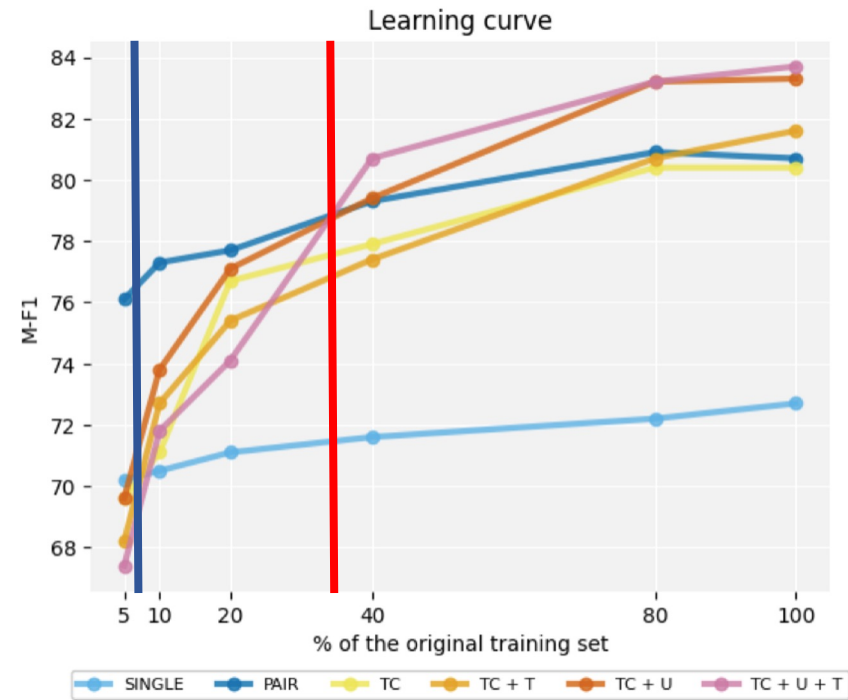
Multi-label classification

Highly unbalanced

Non contextual baselines work better! Why?

Learning Curve Analysis

In **Kialo** Dataset: with **sizes comparable** with the other two datasets, **baselines outperform contextual models**.



Conclusions

Conclusions

We have tested the **effectiveness** of **linguistic** and **extra-linguistic contexts** for text classification.

Full **linguistic context alone** does **not improve** classification performance.

With **extra-linguistic context**, the performance **improves**, especially with the contribution of **structural** context.

Such **results** strongly **depend** on the **amount of data** on which the models are trained.

Extra-linguistic context makes results more **robust** across discussion networks of **different lengths** and **more or less active** users.

Transformer-based models are able to **embed structural features**, which can be effectively given in input to the model in the form of **simple natural language statements**.



<https://github.com/dhfbk/PuCC>



Thanks for
the attention

